

5.5.2 Career Connection in Assistive Tech

معاون ٹیکنالوجی بنانا ایک قابلِ قدر پیشہ ہے۔

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 12 of 12

5.5.2 Career Connection in Assistive Tech

LEARN IT

Designing assistive technology is a **highly rewarding career path** — you earn a living while improving people's lives.

Career	What they do
Assistive Software Developer	Creates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
Biomedical Engineer	Designs artificial limbs and hearing aids.
UX/UI Accessibility Expert	Ensures websites are easy to navigate for visually impaired users (e.g. adding text descriptions to images so screen readers can read them).

UX means *user experience* (how easy and pleasant a product is to use); **UI** means *user interface* (the screens and buttons you interact with). **Accessibility** means designing so that *everyone*, including people with disabilities, can use it.

اردو میں سمجھیں

معاون ٹیکنالوجی بنانا ایک قابلِ قدر پیشہ ہے۔ اسسٹو سافٹ ویئر ڈویلپر اسپیشل ٹو ٹیکسٹ جیسے ٹولز بناتا ہے، ہائیو میڈیکل انجینئر مصنوعی اعضاء اور سماعتی آلات تیار کرتا ہے، اور UX/UI ایکسسبیلٹی ماہر یقینی بناتا ہے کہ ویب سائٹس نابینا افراد کے لیے بھی آسانی سے استعمال ہو سکیں۔

CHECK YOURSELF

1. Who designs artificial limbs and hearing aids?
2. What does a UX/UI accessibility expert ensure?

Answers: 1. A biomedical engineer. 2. That websites are easy to navigate for visually impaired users.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Assistive technology helps:

- a. People with disabilities use systems
- b. Only hackers
- c. Only DBMS keys
- d. Only Agile coaches

MCQ 2. A screen reader is for:

- a. Blind / low-vision users
- b. Faster bubble sort
- c. Wi-Fi
- d. Primary keys

MCQ 3. Captions help:

- a. Deaf / hard-of-hearing users (and many others)
- b. Routers
- c. NAND gates
- d. Pie charts only

MCQ 4. Importance: they support:

- a. Equal access to education and work
- b. Faster CPUs only
- c. Cheaper RAM only
- d. OSI layer 1 only

MCQ 5. A career path is:

- a. Accessibility specialist / AT developer
- b. Only analog radio
- c. Only waterfalls
- d. Only K-maps

MCQ 6. Good HCI + AT means:

- a. Design for a wide range of users from the start
- b. A separate broken version forever
- c. No keyboard needed ever
- d. No contrast ever

Extra short questions

1. Define assistive technology with two examples.
2. Why it matters in a Pakistani college.
3. Link to HCI accessibility principles.
4. One career connection from the book.

Extra long question

1. Propose AT + ordinary design changes so a student with low vision and a student with limited hands can use the college portal. 10 marks. ★ if tagged Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Devices/software that help functional limits: screen readers, switches, captions, magnification.
2. A blind student can still take CS if materials are readable by NVDA and papers have electronic copies.
3. Perceivable, operable, understandable, robust — AT relies on those.
4. AT designer, accessibility tester, special-education technologist.

Long

1. Low vision: contrast, zoom, screen reader labels. Limited hands: large targets, keyboard/switch, extra time. Test with real users. Do not ship a mouse-only form.